MRPL REFINERY - EQUIPMENT ULTRASONIC THICKNESS INSPECTION R

Report ID: IR-2026-P101-04

Equipment Tag: P-101 Transfer Pump

Location: Crude Distillation Unit CDU-101

Inspection Date: 2026-03-15

Connected Process Line: 10-CDU-PL-101

Original wall thickness = 5.0 mm

Current wall thickness = 3.8 mm

Inspection interval = 4 years

Minimum allowable thickness = 3.2 mm

Inspector: Senior PSU NDT Specialist

Status: In Service - Ultrasonic Thickness Survey Complete

PROCESS & INSTRUMENTATION DIAGRAM - CDU-101-01

System: Crude Feed Boost & Transfer Section

Equipment Identified: P-101, T-101

Valves Identified: V-101, V-102, V-103

Connected Process Line: 10-CDU-PL-101

Flow direction = LEFT -> RIGHT

Flow Path Connections: P-101 -> V-101 -> V-102 -> V-103 -> T-101

Isolation Spec: V-101 Suction Gate, V-102 Check Valve, V-103 Discharge Globe

Line Size: 10 inch Sch 40 Carbon Steel

PROCESS & INSTRUMENTATION DIAGRAM - CDU-101-02 AUXILIARY

System: Intermediate Storage & Rundown Section

Equipment Identified: T-101 Storage Vessel

Valves Identified: PSV-101, V-104

Flow direction = TOP -> BOTTOM

Flow Path Connections: T-101 -> V-104 -> Rundown Header

Safety Relief: PSV-101 Set Point 14.5 bar

MRPL STANDARD OPERATING PROCEDURE & MAINTENANCE RULES

Document Ref: SOP-MRPL-MNT-2026

1. Verify latest inspection record and wall thickness prior to overhaul.
2. If current wall thickness is within 0.6 mm of minimum allowable limit, mandatory engineering review requ
3. Verify isolation of V-101, V-102, and V-103 before line entry.
4. Unresolved vibration findings exceeding 7.1 mm/s RMS require immediate trip assessment and mainten
5. Confirm flow direction alignment from P-101 discharge to T-101 storage vessel.

MRPL INSPECTION SIGN-OFF & MATERIAL QA SHEET

Material Specification: ASTM A106 Grade B Carbon Steel

Operating Pressure: 12.5 bar | Design Pressure: 15.0 bar

Operating Temperature: 140 C

Non-Destructive Testing Method: Pulse-Echo Ultrasonic Survey UT

Review Gate: Human Engineering Review Mandatory for Critical High-Risk Operations

Quality Sign-off: Approved by Chief Refinery Integrity Specialist